Aman Singh

AI Software Engineer

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Technical Skills

Languages: JavaScript, TypeScript, Python
Systems & Infra: Linux, Docker, Git, GitHub Actions (CI/CD), REST API design
Access & Security: JWT Authentication, Role-Based Access Control (RBAC)
Automation: Python (cron-based scheduled scripting, DB backup automation)
AI & LLM: LangChain,Langgraph, RAG (Retrieval-Augmented Generation),
Backend: Node.js, Express.js, PostgreSQL, Prisma ORM, MongoDB
Frontend: React.js, Next.js, Tailwind CSS

Experience

Software Developer Intern | *Delhi Skill & Entrepreneurship University (DSEU)* Jun 2025 – oct 2026

- Diagnosed and resolved a production performance issue in grading-scale generation, restructuring PostgreSQL joins/subqueries to cut computation time from **4 minutes to under 300ms** — a systems-level fix, not a feature patch.
- Wrote a **Python automation script** for scheduled daily PostgreSQL database backups via cron, reducing manual DBA overhead and ensuring recoverability of production academic data.
- Built and maintained a role-based **access control (RBAC)** system with JWT authentication, enforcing scoped permissions for faculty across a multi-campus deployment — direct hands-on experience with access management design.
- Built a bulk-messaging platform (Node.js/Nodemailer) with batch processing and account rotation to handle university-wide delivery at scale, including failure handling and retry logic.
- Contributed to **CI/CD pipeline** design using GitHub Actions, automating deployment for academic-workflow REST APIs; collaborated cross-team on operational handoff.

Software Developer Intern | *Penguin Overseas* Jun 2024 – May 2025

- Maintained and enhanced an internal business application by implementing new features, fixing issues, and improving existing workflows.
- Developed inventory and stock management functionalities, enabling efficient product updates and improving operational efficiency.
- Built and integrated REST APIs using Node.js and Express.js with React-based frontend components to support business operations.
- Worked with PostgreSQL to write SQL queries, manage application data, and support deployment, testing, and production maintenance.

Software Development Engineer Intern | *Moreton Bay* May 2024 – Jun 2024

- Collaborated on full-stack development, focusing on frontend UX and backend form-handling logic for a volunteer management system.

Projects

Smart Email Agent GitHub DEMO

- Built an **agentic email assistant** using **LangGraph** and **Groq** that converts natural-language requests into structured email drafts.
- Implemented recipient detail extraction, automated email drafting, and a **human-in-the-loop review workflow** with feedback-driven revisions.
- Integrated **Gmail SMTP** to securely send approved emails, combining LLM-powered reasoning with human oversight.

FrameFusionAI GitHub | Live

- Built a full-stack AI image generation platform using **Next.js**, **Prisma**, and **Google Authentication**, enabling users to generate, store, and share AI-generated images.
- Integrated multiple **Pollinations AI** models, queue-based image generation, API rate limiting, persistent image history, and **Razorpay** payments for premium features.

Education

Bachelor of Technology in Computer Science Engineering 2023 – 2026
G. B. Pant Government College of Engineering, New Delhi